Fuel Charging and Controls

Ford Ecat Electronic Parts Catalog

3.5L Gasoline Turbocharged Direct Injection (GTDI)

WARNING: Do not smoke, carry lighted tobacco or have an open flame of any type when working on or near any fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.

The fuel charging and controls system consists of the:

- fuel injectors.
- fuel injection pump.
- Fuel Pump Control Module (FPCM) .
- fuel rails.
- Throttle Body (TB) .

The fuel charging and controls system is:

- a Gasoline Turbocharged Direct Injection (GTDI) system.
- Pulse Width Modulated (PWM) .

Fuel is metered directly into the combustion chamber. Fuel injectors pulse to follow engine firing order, in accordance with engine demand.

The various sensors detect any changes in the operating conditions and send signals to the PCM. This permits the PCM to control the opening duration (pulse width) of the fuel injectors and maintain optimum exhaust emission control and engine performance for all operating conditions.

Fuel Injectors

NOTICE: Handle the fuel injectors and fuel rail with extreme care to prevent damage to the fuel inlet of the fuel rail, fuel injector sealing areas, and sensitive fuel-metering orifices.

The fuel injectors:

- are electronically operated by the PCM.
- atomize the fuel as the fuel is delivered.
- each have an internal solenoid that opens a needle valve, which injects fuel directly into the combustion chamber.
- are deposit resistant.

Fuel Injection Pump

The fuel injection pump is a single piston and a delivery-on-demand design. The pump is driven from an eccentric on the LH intake camshaft. It can raise fuel pressure from 448 kPa (65 psi) to anywhere between 1,516 and 14,823 kPa (220 and 2,150 psi). The fuel is then supplied to the fuel injectors for distribution into the cylinders.

Fuel Pump Control Module (FPCM)

The FPCM is electronically operated by the PCM and controls the voltage to the FP depending on engine load.

Fuel Rails

NOTICE: Handle the fuel injectors and fuel rail with extreme care to prevent damage to the fuel inlet of the fuel rail, fuel injector sealing areas, and sensitive fuel-metering orifices.

The fuel rails:

- receive and store high pressure fuel from the fuel injection pump.
- deliver fuel to the fuel injectors.

Throttle Body (TB)

The TB :

- controls air supply to the intake manifold by electronically positioning the throttle plate.
- is not adjustable.